Quiz 12 Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Express antiderivatives as functions F with the required derivative. For questions about Riemann integrals, use the definition or the theorem that continuous functions on closed bounded intervals are integrable.

1. Find F on $(-3, 3)$ such that

$$F'(x) = \frac{1}{\sqrt{9 - x^2}}, \qquad F(0) = \frac{\pi}{6}.$$

Verify both conditions and explain why they determine F uniquely.

Solution. For $-3 < x < 3$, define

$$\boxed{F(x) = \arcsin(x/3) + \frac{\pi}{6}.}$$

Since $-1 < x/3 < 1$, the chain rule gives

$$F'(x) = \frac{1/3}{\sqrt{1 - x^2/9}} = \frac{1}{\sqrt{9 - x^2}}.$$

Also, $F(0) = \arcsin 0 + \pi/6 = \pi/6$, so both conditions hold.

To prove uniqueness, let G satisfy the same conditions on $(-3, 3)$. Then $(G - F)' = 0$ on this interval, so the Mean Value Theorem implies that $G - F$ is constant there. Its value at 0 is zero, hence $G = F$ throughout $(-3, 3)$.

2. Let $k \in \mathbb{R}$ and $a < b$. Use the definition of Riemann integrability to prove that the constant function $f(x) = k$ is integrable on $[a, b]$, and determine its integral.

Solution. The function is bounded on $[a, b]$, since $|f(x)| = |k|$. Let $P : a = x_0 < x_1 < \cdots < x_n = b$ be any partition, and choose any sample points $\xi_i \in [x_{i-1}, x_i]$. Then

$$\begin{aligned} S(f; P, \xi) &= \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}) \\ &= k \sum_{i=1}^{n} (x_i - x_{i-1}) = k(b - a). \end{aligned}$$

Set $I = k(b-a)$. Given $\varepsilon > 0$, choose $\delta = 1$. For every partition P with mesh $\|P\| < \delta$ and every choice of sample points,

$$|S(f; P, \xi) - I| = 0 < \varepsilon.$$

This is the definition of Riemann integrability, and therefore

$$\boxed{\int_a^b k \, dx = k(b - a).}$$